FarmShield - Round 2 Evaluation
Solution Architecture & Technical Strategy

Project completion

- Approximately 60-65% production-ready, or 75-80% demonstrable MVP.
- Core crop-recommendation workflow is implemented end to end.
- Calendar, market, storage, and IoT areas are clearly labelled prototypes/demos.

1. Architecture

React + TypeScript frontend -> FastAPI backend -> ML model + cleaned dataset

- Frontend: React 19, TypeScript, Vite dashboard.
- Backend: Python FastAPI REST API with validated inputs.
- Data/ML: CSV dataset, cleaning pipeline, persisted joblib model.
- APIs: health, metrics, data summary/preview, and crop prediction.

2. Technical Feasibility

- Farmers provide N, P, K, temperature, humidity, pH, and rainfall.
- The API validates input ranges and returns five ranked crop recommendations.
- The trained model is saved and reloadable; it is not retrained per request.
- Cleaning removes invalid/duplicate records and excludes yield/price leakage.
- The dashboard connects directly to the live local prediction API.

3. Technology Choice

Component	Technology	Purpose
Frontend	React + TypeScript + Vite	Responsive, typed dashboard
Backend	FastAPI	Fast API and input validation
ML/Data	Python, pandas, sklearn	Tabular data processing/modeling
Model storage	joblib	Persist/load sklearn pipeline
AI algorithm	Random Forest Classifier	Nonlinear tabular classification

4. Data / AI / Algorithm Suitability

Deployed model: RandomForestClassifier \(300 decision trees\)

Problem: Multi-class crop recommendation

Dataset: 2,200 records; 22 crop categories; 7 input features

Features: N, P, K, temperature, humidity, pH, rainfall

Training/testing: Stratified 80/20 holdout split \(random_state=42\)

Training records: 1,760 Test records: 440

Evaluation metrics

Accuracy: 99.32%

Macro Precision: 99.35%

Macro Recall: 99.32%

Macro F1-score: 99.32%

R2 score: Not applicable - this is classification, not regression.

Important: This performance is on a held-out split of the supplied dataset.

It is not evidence of field accuracy; external field validation is still needed.

5. Implementation Scope

Completed

- Crop recommendation ML pipeline, data cleaning, validation, and model artifact.
- FastAPI prediction API and frontend-to-backend integration.
- Top-five crop recommendations with model probability scores.
- Dataset transparency/metrics endpoints and dashboard support.
- Crop calendar UI, harvest/storage calculator, and provider management UI.
- Local browser persistence for the demonstration workflow.

Prototype / demo only

- Crop calendar uses rule-based day counts; DSSAT is not connected.
- Market prices/history, storage listings, and IoT readings are sample data.
- No live weather, verified inventory, real booking, payments, or live GPS.
- No production authentication, cloud database, or user-management workflow.

Notebook status

- Cropcalendar.ipynb demonstrates GDD/rule logic and historical-weather experiments.
- It is not integrated as a validated production crop-calendar model.
- Befirecropmodel.ipynb contains an earlier suitability-model experiment; the deployed application model is the RandomForestClassifier in backend/train_model.py.

6. Risks and Contingencies

Risk: Dataset provenance is undocumented.

Mitigation: use government/university data; record geography, season, source, and units.

Risk: Very high offline score may not generalize.

Mitigation: cross-validation plus independent regional and field test sets.

Risk: No real-time weather or validated crop simulation.

Mitigation: integrate trusted weather data; add DSSAT only with validated soil,
cultivar, weather, and farm-management inputs.

Risk: Market, storage, and IoT information is simulated.

Mitigation: integrate verified mandi data, warehouse partners, and sensor ingestion.

Risk: No production security/data layer.

Mitigation: add authentication, PostgreSQL, encrypted secrets, backups, and audit logs.

Presentation conclusion

FarmShield has completed its core AI-enabled crop-recommendation MVP. The next
phase is verified data integration, external/field validation, live services,
and production security before making farm-critical recommendations.